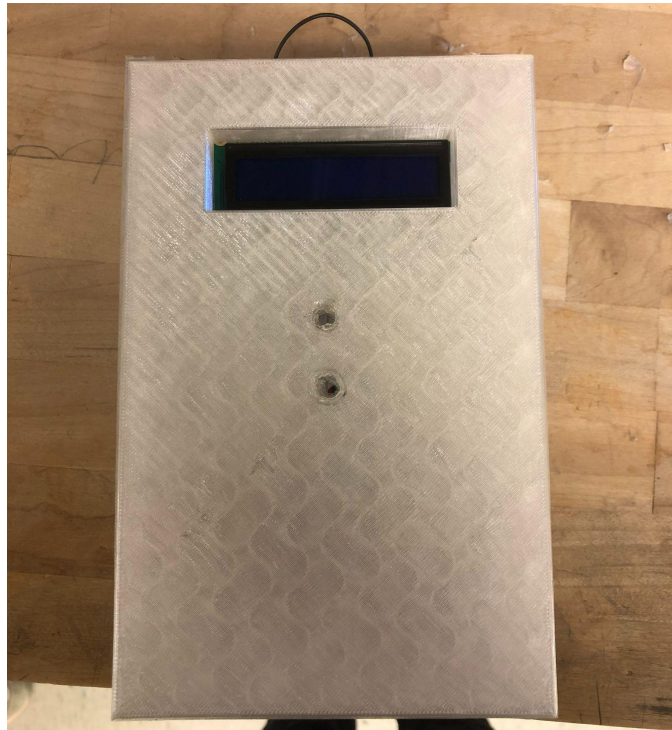


Solar Powered Emergency Beacon



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Table of Contents

Table of Contents	ii
Tables and Figures	iii
Tables	iii
Figures	iii
Executive Summary	iv
1. Introduction	1
1.1 Background Information	1
1.2 Analysis of Existing Solutions	1
1.3 Introduction to the Device	2
2. Overview of Prototype	2
2.1 The Emergency Beacon	3
2.2 Prototype Schematic	3
2.3 Subsystems of the Emergency Beacon	4
3. Testing	5
3.1 GPS	6
3.2 Bluetooth	6
3.3 Display	6
3.4 Solar Panel	7
4. Emergency Beacon Prototype Analysis	7
4.1 Requirement Validation	8
4.2 Prototype Design Tradeoffs	8
4.3 Possible Emergency Beacon Improvements	8
4.4 COSPAS-SARSAT Integration	9
5. Value Proposition	10
5.1 Stakeholder Analysis	10
5.2 Cost Analysis	11
5.3 Recommendation	12
6. Conclusion	12
References	14
Appendix	17

Tables and Figures

Tables

Table 1: Project Requirements vs. Prototype Capabilities	11
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Figures

Figure 1: Emergency Beacon Circuit Overview	6
Figure 2: Emergency Beacon Component Schematic	7
Figure 3: COSPAS-SARSAT Infrastructure Overview [20]	13

Executive Summary

The increasing popularity of hiking in remote locations has led to a higher number of accidents and deaths. This report exists to explore the development of a solar-powered emergency beacon prototype that addresses the shortcomings of existing solutions and provides utility to hikers in emergency situations.

Current Personal Locator Beacons (PLBs) are typically single-use and require external power sources, which can be dangerous during extended periods in remote areas. There is a need for a multi-use and rechargeable solution to improve safety and reassurance for outdoor enthusiasts.

Our team developed a solar-powered emergency beacon prototype that is capable of multiple uses. The device requests GPS data, displays it to the user, and transmits it to a receiving station, all while being powered by a solar panel. The prototype underwent a series of tests to evaluate its functionality, efficiency, and viability as a commercial product. Key aspects of testing included the evaluation of the GPS module, Bluetooth communication, solar panel charging capabilities, and overall performance in meeting project goals.

The initial prototype successfully met most project goals, but testing revealed some flaws that could persist into the future. Upgraded components for future prototype versions could lead to higher costs and increased weight, making the device less practical for hikers looking to carry the device for an extended period of time.

This report details the design and development process, prototype testing, and recommendations for further research and investment. Ultimately, the report concludes that it is not feasible to create a cost-effective, powerful, and portable solar-powered emergency beacon with current technology.

1. Introduction

This section will include background information concerning our problem along with an analysis of existing solutions and an introduction to our unique solution.

1.1 Background Information

Hiking is an extremely popular hobby for many outdoors enthusiasts, and for good reason. It's a great way to combine exercise with exploration, and it can be extremely peaceful to be alone with nature. However, it is more dangerous than you may think. There are typically around a thousand deaths every year in national parks. 50% of these are attributed to unintentional causes such as falls, drowning, or straying off trails [1]. There has been a rise in the popularity of outdoor recreation, this has led to an increase in hiking-related accidents and death by 21% [1]. Hiking is a getaway for many people, especially in recent times, with many indoor recreational activities being shut down due to pandemic protocols. With hiking becoming increasingly popular, accidents and missing people are bound to increase as well.

Tragically, many people go missing in national parks on an annual basis [2]. Not only is the well-being of the hiker at risk, but the families have to go through both the grief of a loved one becoming injured or even dying in addition to other issues presented when the person is not found. For example, in the state of California, it can take five years to receive a death certificate for a missing person [3]. This means that the family can't collect the missing person's life insurance or social security which is usually used to help the family recuperate after losing their loved one.

When someone does go missing on a trail, it can seem impossible to recover them. Yosemite National Park has 750 miles of trails, which makes it a nightmare when it comes to locating missing or endangered hikers [3]. Without a way to track the hiker's location, emergency responders have to haphazardly scan the vast wilderness, sometimes to no avail. This problem needs to be addressed in order to increase the safety of hiking and save the lives of those who experience incidents on trails.

1.2 Analysis of Existing Solutions

This problem has several existing solutions, but all of them prove to be ineffective when faced with certain scenarios. A popular "solution" to the problem is cell phones. Many people believe that they can use their phone to signal for help when presented with an emergency in the wilderness. However, there are several reasons why a cell phone is an ineffective emergency beacon for distressed hikers. Many national parks restrict cell towers to a few limited locations in order to protect park resources. [4] Cell phones also have limited battery capacity, with the majority of them lasting between 10 and 17

hours. [5] This is not enough for people who are on longer hikes, or forgot to fully charge their phone. So, cell phones do little to combat the problem of missing hikers. There are alternative solutions, such as Personal Locator Beacons (PLB) and Satellite Messengers. These devices have a much longer battery life than phones and can get better service; however, these devices are often single-use [6]. It is inefficient to replace devices one time after use. Also, if your trip takes two unfortunate turns, the single-use device is useless. Finally, these devices always have a high price tag, which the average hiker may not be willing to pay for [6]. For these reasons, the multi-use, solar powered device that we are producing could seem very appealing to the primary stakeholders, hikers.

1.3 Introduction to the Device

Difficult problems require unique solutions. In order to combat the disappearance of hikers in National Parks, research was conducted to design a new, more effective PLB.

The Solderers built a prototype device that could potentially save many lives. The device takes the concept of a PLB one step further by introducing solar power as the main energy source. The prototype device, while lacking the proper components to be completely effective, was a proof of concept that it is possible to construct a solar powered emergency beacon. The aspect of solar power brings a layer of safety and reliability that no other PLB on the market can boast. The prototype is also multi-use, opposed to battery powered PLBs which can only be used once before the battery must be replaced. The rest of this document will elaborate on the design and testing processes, before evaluating the effectiveness of the experimental product.

2. Overview of Prototype

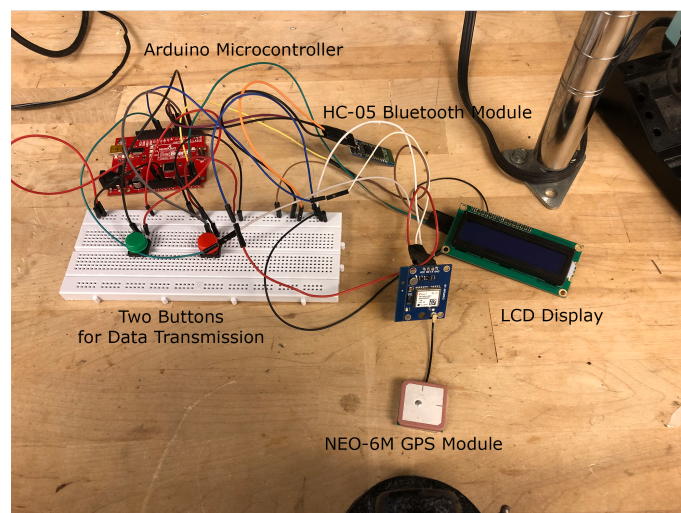


Figure 1: Emergency Beacon Circuit Overview

2.1 The Emergency Beacon

Personal Location Beacons or emergency beacons provide safety and reassurance to every hiker. It was important to capture the essential uses for the first prototype of the beacon.

The essentials of emergency beacons include retrieving data to the device, sending data to a receiving station, and an unwavering battery life. The first prototype of our emergency beacon was able to accomplish all of these.

Where our emergency beacon differs is the solar panel for recharging. Usually, emergency beacons are single use. However, our goal was to create an emergency beacon that can be used multiple times. This is possible thanks to the solar panel. The solar panel recharges the device automatically. This is a necessity in the wild, where there are no outlets.

2.2 Prototype Schematic

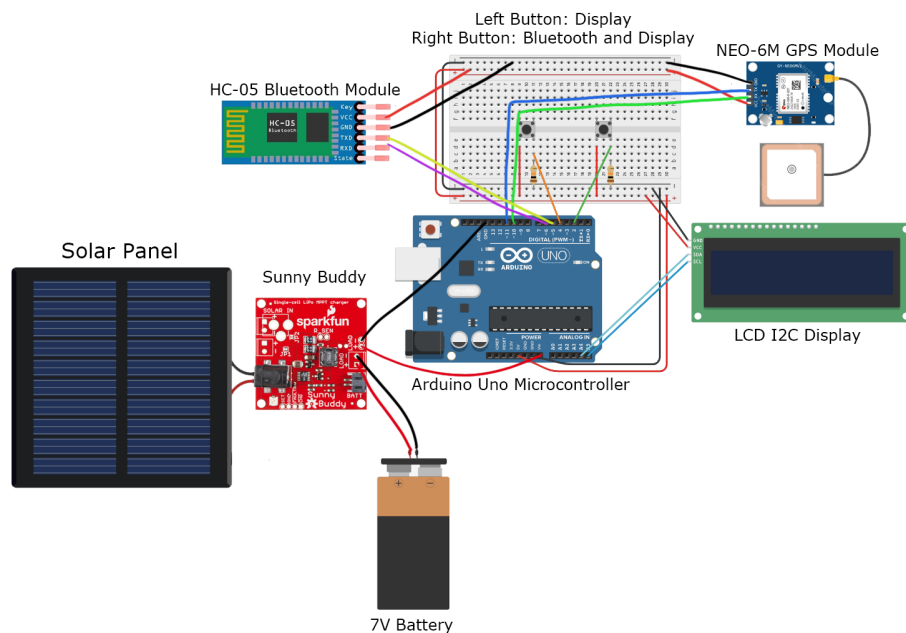


Figure 2: Emergency Beacon Component Schematic

The schematic shows how each component is connected. While the wires connect the circuit, the Arduino microcontroller actually connects each component to each other. You can split the circuit up into two ideas, inputs and outputs.

The inputs include the Sunny Buddy and the NEO-6M GPS module. As explained earlier, the Sunny Buddy inputs power into the circuit either through the battery or the solar panel. This depends on the amount of sunlight available at the time. The NEO-6M inputs raw GPS data into the Arduino.

The outputs include the HC-05 bluetooth module and LCD display. They are powered by the Sunny Buddy input. The LCD display outputs the NEO-6M GPS data to the user as a screen. The HC-05 displays the output to a receiving station as an output.

The buttons call for the NEO-6M to run. Depending on the button, the data is displayed to one or both of the outputs.

2.3 Subsystems of the Emergency Beacon

2.3.1 Powering the Device

The emergency beacon relies on an Arduino Microcontroller as its centerpiece. An Arduino is a “plug and play” device, whose analog pins allow for data tracking as an input and output. Arduino is referred to as “plug and play” as it is simple to connect modules and simultaneously write software.

A 9 volt rechargeable battery will be charged by the solar panel to use as stored power. In cases where there is not enough sunlight, or the device is not in use, it is important to have an alternative power source constantly charged. The 9V battery solves this issue.

The Sunny Buddy connects the solar panel, battery, and Arduino together. The Sunny Buddy will ensure the energy collected by the solar panel is being applied appropriately. If the irradiance is high, the Sunny Buddy will allow the solar panel to power the Arduino and charge the battery. If the irradiance is low, the Sunny Buddy lets the battery power the Arduino.

2.3.2 Arduino Modules

The location of the user will be found using a NEO-6M GPS module. GPS stands for Global Positioning System. The GPS module will retrieve its longitude and latitude coordinates. It will also return the date and time. When the GPS module is on and returning coordinates, it takes 0.225 W of power [7].

The data will be displayed to the user by a small LCD display. The LCD display uses 0.005 W of power [8]. The size of the LCD display allows for 2 lines of 16 characters for 32 total characters.

An HC-05 bluetooth module is the first prototype's method of data transferring. The HC-05 module requires 0.04 W of power in use [9]. An Android phone is the first prototype's method of a receiving station. In reality, radio waves sent to a satellite would send the GPS data to an emergency response team. Since we do not have access to government satellites, we used the easiest form of wireless data transfer which is bluetooth. The HC-05 connected to the Arduino sends the GPS module's data to the Android once connected.

The Android receiving station has an application which displays information such as: coordinates, date sent, time sent, and the user.

2.3.3 Power Control

All of these modules do not run constantly. If that were the case, the power output would be too large to maintain a charge. To ensure this does not happen, two buttons are in place to control the subsystems. Both buttons request NEO-6M data, however, the destination of the data is different.

The first button will show the NEO-6M data to the user via the display. If the user is lost, they can compare where they are to an atlas. They can simply use it as a clock as well. Another form of confirmation for location is always necessary for reassurance.

The second button will send the same NEO-6M data to the display, but in addition will send the data to the receiving station. In a real scenario, the receiving station is an emergency response team. This is for scenarios where the user is in danger and needs help.

The total power consumed is equal to 0.27 W. This is less than the amount the solar panel supplies on a day with 8 hours of sunlight, 0.426 W. This number was found through testing.

3. Testing

Our device uses several components: GPS, Bluetooth, a display, and a solar panel. After putting these components together, we ran a test on each to see if it met up to the standards of our technical requirements. The code we ran on our beacon can be found in appendix A.

3.1 GPS

3.1.1 Test

Our technical requirements said that we wanted our user's coordinate location to be accurate up to 2.5 m [7]. To test this, we would compare the GPS module's coordinate data to the GPS data that I pull from my phone at the same location. This will give us some difference in latitude and longitude which we will convert to meters.

3.1.2 Result

According to my phone, I was standing at (42.293466°, -83.711022°). According to our GPS module, I was standing at (42.293401°, -83.711182°). This means that there was a difference in latitude of $6.5e-5^\circ$ and longitude of $1.6e-4^\circ$. After doing some conversions, this corresponds to a linear distance of 19.193612 meters.

3.2 Bluetooth

3.2.1 Test

When we pitched our project, we sought that our Bluetooth module would send data up to 10 meters [10]. To test this, Garrett will click the button to send GPS data to my phone via Bluetooth. The data also contains a timestamp that includes seconds, so I will keep track of whether or not the data updates based on whether or not time updates. Then, I will walk five steps, wait for Garrett to press the button, and I will confirm whether or not I received the new data. Then, we will measure how far I walked from the beacon.

3.2.2 Results

After walking just over 60 meters, I reached a wall where I continued to receive new data sent by Bluetooth. Having reached the wall where I could not continue forward to discover its actual range limit, we decided to conclude the test. This result is incredibly shocking because it exceeds three times the range from the part's specifications [10]; nevertheless, we conclude that our Bluetooth can send data 60+ meters.

3.3 Display

3.3.1 Test

A test for our display is simple. We want to display our GPS's location, date, and time data on screen for several seconds on a digital screen. After the screen displays the data for several seconds, it should clear the data and turn off until it is triggered by another button press, at which point it should display the updated data.

3.3.2 Result

After soldering in the pins for our display, it worked as intended. When a button is

pressed, the digital screen turns on and shows the beacon's latitude and longitude as well as the date and time for seven seconds.

3.4 Solar Panel

3.4.1 Test

We need to test how effective our solar panel is at powering our beacon and charging our battery. To do this, we will place our solar panel under a sunlamp and measure its wattage. With this wattage and the wattage that our beacon consumes, we can calculate the charge and discharge time of the battery.

3.4.2 Result

After testing, we measured the wattage of our solar panel to be 0.426 W. According to the specifications of our components, the GPS uses 0.225 W, the Bluetooth uses 0.04 W, and the display uses 0.005 W [7] [9] [8]. This means that the total power our beacon consumes is 0.27 W. Connected to our 9.6 Wh battery, it will take 22.5 hours to charge our battery from zero if our device is not left running. If the beacon is on and is not being supplied with any power from the solar panel, it will last 35.6 hours starting from a full charge.

4. Emergency Beacon Prototype Analysis

The testing section demonstrates the effectiveness of the beacon's functions. We can now investigate the extent to which the beacon meets the original project requirements.

Table 1: Project Requirements vs. Prototype Capabilities

Project Requirements	Prototype Capabilities
Pair bluetooth module with mobile device and transmit data, up to 30 ft.	Pairs bluetooth module with mobile device and transmits data, up to 200 ft.
Coordinate location accurate to 2.5m.	Coordinate location accurate to 19.19m.
Display user's coordinates on digital screen.	Displays user's coordinates on digital screen.
Have buttons for receiving and transmitting data.	Has buttons for receiving and transmitting data.

4.1 Requirement Validation

The emergency beacon prototype was able to meet most of our project goals and specifications for the project. It was able to pair the bluetooth module with a receiving device and transmit data farther than the initial 30 ft prediction, up to 200+ ft. It was able to display the coordinates of the user upon button press and the GPS module provided data accurate to 19.19m. This error exceeds the original goal of 2.5m but is still reasonably accurate. The 9V solar panel was also able to charge the 9.6Wh battery within less than a full day and it provided enough power to run all components when in direct sunlight.

4.2 Prototype Design Tradeoffs

The beacon prototype was successful in completing our objectives, but its completion came with some drawbacks. Our team had to prioritize the circuit and component success over the aesthetics of the design due to insufficient time. We didn't iterate through enough tests of the aesthetic design of the prototype in order to get everything working within the PLA plastic housing. Our CAD design failed to account for the placement of the buttons and the pins on the display. This meant we were unable to make the buttons function properly through the hand-made holes in the plastic shell and the 16x2 LCD display became a difficult fit. We instead left the prototype open circuit with the buttons attached directly to the breadboard, which was secured inside the PLA plastic housing.

Another trade-off was the display, as we initially planned on using a higher resolution display. The original display proved difficult to integrate with the Arduino and other components, so we switched to a display with a 32 character limit because it was easier to test. The 16x2 LCD display limits the precision of the coordinate output to 2 decimal places, which equates to a 1.1km radius [11]. This drawback prevents the user from accessing the full 4 decimal place precision, which equates to a 11.1m radius [11].

These trade-offs were purely aesthetic, as they did not prevent the prototype from performing any of its intended functions.

4.3 Possible Emergency Beacon Improvements

The current beacon prototype provides proof of concept, but future iterations of the design will require improved features to ensure hiker safety in dire circumstances. The most important change would be replacing the bluetooth module with the COSPAS-SARSAT 406MHz Module and 406MHz antenna [12]. This would allow the device to emit the 406MHz SOS signal to be picked up by the SARSAT satellite system

[13]. This single module improves the beacon's range from roughly 200 ft to global coverage, which is a necessity considering the device is only needed when the user is outside of cell-coverage [13].

The 406MHz signal needs a 5W signal to compete with other beacons on the market and ensure it's received through any foliage [14]. This higher wattage signal requires a much stronger battery than the 9.6Wh battery used in the first prototype. A 120 Wh battery is capable of emitting the 5W signal for the COSPAS-SARSAT required time of 24 hours [15]. The new battery would then need a solar panel capable of producing more power, and preferably one that is smaller. The current solar panel's size of 19.5cm x 12.5cm limits the device's ability to be handheld, so a more compact 11.5cm x 11.5cm 11V solar panel would be an ideal replacement [16].

A future beacon prototype would also need to be highly resistant to the elements, such as being IPX7 water & IP6X dust proof [17]. This could be accomplished by replacing the PLA plastic housing with weatherproof acrylic plastic [18]. Additionally, the user would need access to the most precise coordinates possible. The 128x64 OLED is a potential replacement, as an upgraded display would provide space for full 4 decimal point precision [19].

These improvements would come with some drawbacks, as the new components would increase the cost and weight of the device. These sacrifices are necessary in order for the device to provide its intended utility.

4.4 COSPAS-SARSAT Integration

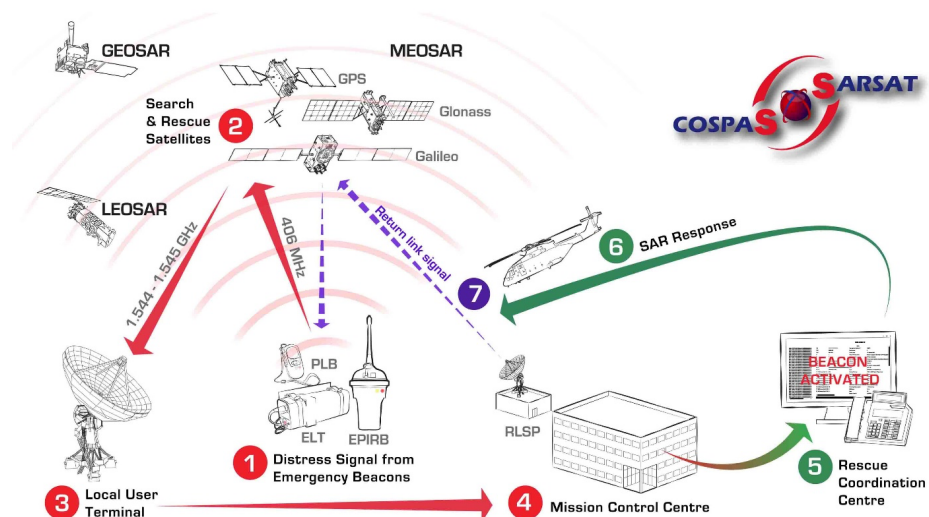


Figure 3: COSPAS-SARSAT Infrastructure Overview [20]

The reason future prototypes would use the COSPAS-SARSAT 406MHz Module instead of other satellite communication components would be to make use of the previously existing infrastructure seen in Figure 3. There are many moving parts involved between the SOS signal being sent and the user being rescued. The signal is initially picked up by ground stations referred to as local user terminals, which then processes the data and communicates the beacon's location to a mission control center (MCC). After the MCC validates the signal they notify the nearest search and rescue response team [20].

Using another component to communicate to a different satellite system would require our own receiving station and validation procedure, making the initial investment exponentially more expensive. This means that any future prototypes must meet COSPAS-SARSAT requirements and pass their testing procedure in order to gain the COSPAS-SARSAT Type Approval Certificate [21]. The device must then be registered with the National Oceanic and Atmospheric Administration (NOAA) by any potential user as required by law [22]. COSPAS-SARSAT is a necessary partnership to the design and testing of future prototypes.

5. Value Proposition

This section will break down why our beacon is useful to our stakeholders, analyze production cost and provide a recommendation for further research and investment in our product.

5.1 Stakeholder Analysis

People go missing for many reasons, but the inability to seek help from others is always a reason why people cannot be found. "Cold cases", as they are called, are recorded back to the 1950s [2]. Going missing is nothing new, but the difference in technology between the 1950s and today is extraordinary. In a world full of evolving technology, why hasn't the issue of missing hikers been resolved? Daryl Miller, a retired mountaineering ranger in Alaska believes, "Technology has served to blunt respect for the tests Mother Nature can still throw at humans" [23]. The amount of technology today has blinded hikers from the harsh reality of nature. People are too arrogant in nature, and a lack of safety can cause unfortunate outcomes. However, we believe there is a world where technology can be used in nature to create safety.

Many adventurers like to explore nature to get away from the internet and their technologies. However, it is this technology that could be the difference between life and death. When deep in nature, you are considered "off the grid," referring to the electrical grid. Cellular devices usually have no internet or service. Nothing can be sent or received. Even if you could get a signal, your phone has a limited battery life. Some

may be planning on just hiking for the day and think that a portable charger may be enough safety. If plans go awry, the day trip could turn into weeks.

While Hikers will be the primary audience for marketing, anyone who spends a large amount of time in remote outdoor locations will find the beacon useful. The product could be marketed through various means, and mutually beneficial partnerships could be sought out with companies such as REI Co-op, who sell a large amount of hiking gear in their stores. The solar powered beacon will provide them with an opportunity to sell a unique product, while providing the hikers exposure and easy access to the beacon. Another company who's partnership would serve as valuable is Garmin. They are a current leading seller of GPS devices, and the solar powered beacon could prove useful to them for similar reasons as REI Co-op. Seeking out these partnerships could serve to advance the production and distribution of solar powered distress beacons by providing a means to spur widespread sale of the product and eventually adoption of the product into hiking culture.

5.2 Cost Analysis

There are several components of cost that will need to be accounted for in the production and distribution of the product. The first portion to analyze is the cost of parts. The component prices will be overestimated. Public market prices will be used to represent the costs, but a different price would be set for mass orders to an established company. The real-world product will need a neo-6M GPS module, a microcontroller, a display, a small solar panel and a 120 Wh rechargeable battery. It will also need a specific component that only SARSAT can supply, due to their monopolization on 406MHz frequencies. For the sake of simplicity, assume that this component costs 5 dollars, as that is the price of similar available components such as a 433MHz transmitter module. Adding up the remaining prices gives \$198.5 in component costs. (u)

The next cost to analyze is that of assembly. Assume that a factory has already been purchased using seed money from investors. A reasonable wage for the factory workers would be around \$12 an hour. (v) There will be 10 workers, each making 6 units per hour and a manager who makes \$30 an hour. (w) 60 units will be produced per hour, with a total cost of \$150 an hour in wages. This results in \$2.5 per unit in assembly costs.

Lastly, distribution must be taken into account. Ideally, the product should be available to anyone, from a hobby hiker to a hardcore mountaineer, so the sales will be done through an online website. It is a necessity to be able to ship beacons from the factory right to the user. Shipping a medium sized box to another state through a medium such as a postal service typically costs around \$10.(x) In total, a reasonable estimated value

for a singular unit is \$211, taking into account the component costs, as well as assembly and distribution. A reasonable price tag for the customer is \$300, lower than many battery powered personal locator beacons. This results with in an impressive profit margin, slightly over 30%

5.3 Recommendation

The product provides clear utility to the primary stakeholders by providing a layer of safety and efficiency to the user that non-solar powered products cannot. It does so without sacrificing important features such as accurate coordinate location, weatherproofing, partnership with SARSAT, and data display on a digital screen. Another benefit the solar panels provide is the ability to run the display and collect GPS data without draining the power supply needed to send the radio signal.

While there are no other solar powered distress beacons on the market, the addition of a rechargeable battery comes with some drawbacks. It was discovered through testing and research that a 120 Wh battery would take approximately 10 days to completely charge via 9V solar panel. This prohibits the prospect of sending multiple full length signals, but there could still be some benefit to partially charging the battery and sending a shorter length signal. This could be useful in the event of an evacuation, where the user's location continuously changes.

Another major drawback of the solar powered version is its weight. Portability is a vital feature, as it must be able to be carried through tough trails over long spans of time without being a hindrance to the user. Rechargeable batteries that store the power requirement of 120 Wh are much heavier than the non-rechargeable batteries. This will cause the commercialized beacon to weigh around 4 lbs, whereas an average PLB weighs around ½ lbs. (y).

For these reasons, with current technology it is irresponsible to recommend further investment and research in the product. In order to move forward with the beacon, some new technology would be required. Examples include advanced solar technology that allows more charge to be produced, and a lighter model of the 120 wH rechargeable battery.

6. Conclusion

As outdoor recreation continues to increase, so do accidents and deaths. In some cases, these accidents can be avoided with Personal Locator Beacons.

The Solderers created a unique prototype not seen on the market. Where most Personal Locator Beacons are single use, our product is not. The solar panel creates a

multi-use solar powered emergency beacon that adds another layer of reassurance to hikers. In the case an original signal was not received, or an updated location needs to be tracked; the solar powered beacon shows its uniqueness.

The first rendition included a series of components that can be defined as inputs and outputs, leading into an Arduino microcontroller. The Sunny Buddy combines with the solar panel and battery to input power to the device. The NEO-6M module brings in raw GPS data. Select data is then displayed to an LCD display or sent to a receiving station by a HC-05 bluetooth module.

Through testing of the first prototype, flaws were found. The amount of time it took to charge the device makes it inefficient. However, upgrading these pieces in a future prototype led to worse conclusions. The price was above market standards and the weight made it not handheld.

Every goal set was achieved in the first prototype. The device can request GPS data, display this data to the user, send it to a receiving station, and do so by being charged entirely by a solar panel system. No trade-offs were necessary for the first prototype, however, many ideas came to mind on how an upgraded model would look for version two. A second version would not be pursued, though.

After testing, it was clear that the idea could not be recommended. With current technology, it was not possible to create a cheap, powerful, handheld product. This was demonstrated clearly through a market analysis for our target customer, the everyday hiker.

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Appendix A: Personal Locator Beacon Code

```
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>
#include <TinyGPS++.h>

// BT
static const uint32_t BTBaud = 115200;
static int const BTRXPin = 5, BTTXPin = 6;
SoftwareSerial BTserial(BTRXPin, BTTXPin); // RX | TX

// GPS
TinyGPSPlus gps; // The TinyGPS++ object
static const uint32_t GPSBaud = 9600;
static int const GPSRXPin = 11, GPSTXPin = 10;
SoftwareSerial ss(GPSRXPin, GPSTXPin); // The serial connection to the GPS device

// Display
LiquidCrystal_I2C lcd(0x27, 16, 2); // I2C address 0x27, 16 column and 2 rows

// number of decimals
static const uint32_t numDecimalsDisplay = 2;
static const uint32_t numDecimalsReceiver = 5;

// Buttons
int buttonOne = 2; // set buttonOne to pin
int buttonTwo = 4; // set buttonTwo to pin
int buttonOneState = LOW;
int buttonTwoState = LOW;

void setup() {
    BTserial.begin(BTBaud);
    ss.begin(GPSBaud);
    lcd.init(); // initialize the lcd
    lcd.backlight();
    pinMode(buttonOne, INPUT);
```

```

pinMode(buttonTwo, INPUT);
lcd.clear();          // clear display
}

void loop() {
  buttonOneState = digitalRead(buttonOne);
  buttonTwoState = digitalRead(buttonTwo);
  while (buttonOneState == HIGH && ss.available() > 0) {
    gps.encode(ss.read());
    if (gps.location.isUpdated()) {
      // Latitude in degrees (double)
      lcd.setCursor(0,0);
      lcd.print("LAT");
      lcd.setCursor(4,0);
      lcd.print(gps.location.lat(), numDecimalsDisplay);

      // Longitude in degrees (double)
      lcd.setCursor(0,1);
      lcd.print("LON");
      lcd.setCursor(4,1);
      lcd.print(gps.location.lng(), numDecimalsDisplay);

      // Month (1-12) (u8)
      lcd.setCursor(11,0);
      lcd.print(gps.date.month());
      lcd.print("/");
      // Day (1-31) (u8)
      lcd.print(gps.date.day());

      // Hour (0-23) (u8)
      lcd.setCursor(11,1);
      lcd.print(gps.time.hour());
      lcd.print(":");
      // Minute (0-59) (u8)
      if (gps.time.minute() < 10){
        lcd.print(0);

```

```

    }
    lcd.print(gps.time.minute());

    delay(7000); //delay before clearing
    lcd.clear(); //clear the display
}
}

while (buttonTwoState == HIGH && ss.available() > 0) {
    gps.encode(ss.read());
    if (gps.location.isUpdated()) {
        // Latitude in degrees (double)
        lcd.setCursor(0,0);
        lcd.print("LAT");
        lcd.setCursor(4,0);
        lcd.print(gps.location.lat(), numDecimalsDisplay);
        BTserial.print(gps.location.lat(), numDecimalsReceiver);
        BTserial.print(",");

        // Longitude in degrees (double)
        lcd.setCursor(0,1);
        lcd.print("LON");
        lcd.setCursor(4,1);
        lcd.print(gps.location.lng(), numDecimalsDisplay);
        BTserial.print(gps.location.lng(), numDecimalsReceiver);
        BTserial.print(",");

        // Year (2000+) (u16)
        BTserial.print(gps.date.year());
        BTserial.print(",");
        // Month (1-12) (u8)
        lcd.setCursor(11,0);
        lcd.print(gps.date.month());
        lcd.print("/");
        BTserial.print(gps.date.month());
        BTserial.print(",");

```

```

// Day (1-31) (u8)
lcd.print(gps.date.day());
BTserial.print(gps.date.day());
BTserial.print(",");

// Hour (0-23) (u8)
lcd.setCursor(11,1);
lcd.print(gps.time.hour());
lcd.print(":");
BTserial.print(gps.time.hour());
BTserial.print(",");
// Minute (0-59) (u8)
if (gps.time.minute()<10){
  lcd.print(0);
}
lcd.print(gps.time.minute());
BTserial.print(gps.time.minute());
BTserial.print(",");
// Second (0-59) (u8)
BTserial.print(gps.time.second());
BTserial.print(";");

delay(7000); //delay before clearing
lcd.clear(); //clear the display
}
}
}

```